

Math (Grade 1, Semester 1)

Right Triangle Trigonometry

Weeks: Week 01 - Week 04

MA.1.01:

HSS MA.1.01 TILL MA.1.11

Create, interpret and analyze trigonometric ratios that model real-world situations.

Concepts	Skills
•1. Unit circle •2. Six trigonometric functions (without graph the function) •3. Related angles •4. Sine law, cos law •5. Angle of elevation and angle of depression 6- Directed angle (added) 7- Positive, negative, and co-terminal angles of rotation (moved from MA 1.09)	•1. Sine, cosine, and tangent ratios and Pythagorean Theorem •2. Define the six trigonometric functions in terms of right triangles. •3. Determine which quadrants have positive and negative trigonometric values •4. Derive sine and cosine law •5. Solving triangles (using law of sine and cosine) •6. Use the law of sines, law of cosines to solve applied problems. •7. Use the inverses of the sine, cosine, and tangent to solve problems. 8-Relate the coordinates of points on the unit circle to the sine and cosine of angles •10. Draw and identify positive, negative, and co-terminal angles of rotation

Essential Question(s): •"Why is modeling an indispensable tool?"

•Why do we represent phenomena in multiple ways?"

Big Idea:

We use mathematical modeling with a variety of classes of functions to solve applications such as in biology and finance, and to form a basis for the study of mechanics and calculus.

Textbook and Resource Materials:

Math Connections 2a sections 1.8, 3.1 - 3.4 , 3.6, 3.7

See materials in Unit Materials folder:

<https://drive.google.com/drive/folders/0B9KK7ZY6QeS3dGIETUVNQ0c2em8>

Evidence of Learning:

Problem Set 1.8 - BA #3, ST #6d

Problem Set 3.1 - BA #6, 10, ST #9

Problem Set 3.2 - BA #5, ST #7c

Problem Set 3.3 - BA #7, ST #8

Problem Set 3.6 - BA #1, ST # 8

Capstone Connection:

Determine the dimensions of the sustainable houses (**dimensions of any geometrical shapes**).

The Basics Of Statistics

Weeks: Week 5 - Week 6

MA.1.02:

HSS MA.5.0

Analyze, display and describe quantitative data with a focus on standard deviation.

Concepts	Skills
•1. Shape, center, spread 2-. Histogram 3-. Stem plot •4. Dot plot 5. Skewed •6. Symmetric 7. IQR 8. Boxplot 9-. Outliers 10-Scatter plot 11-- Correlation 12-. Correlation coefficient (13-Regression line 14. Standard deviation (without frequency)	•1. Describe the overall pattern (shape, center, spread) of a distribution and identify any major departures from the pattern (like outliers). •2. Make a histogram, dot plot, boxplot or stem plot to display small sets of data and interpret the data. •3. Identify the shape of a distribution (symmetric or skewed) from a dot plot, stem plot, or histograms •4. Calculate and interpret measures of center (mean, median) and spread (IQR, standard deviation). •5. Use appropriate graphs and numerical summaries to compare distributions of quantitative variables. 6- calculate correlation coefficient (Spearman's rank and Pearson's linear correlation coefficient) 7- find the equation of regression line 8- calculate the value of y given x using regression equation

Essential Question(s): •How can graphical displays be manipulated to present misleading information?

- How can data analysis be used to predict future happenings?
- Does the data always lead to the truth? Why or why not?

Big Idea:

We use mathematical modeling with a variety of classes of functions to solve applications such as in biology and finance, and to form a basis for the study of mechanics and calculus.

Textbook and Resource Materials:

Starnes, Daren - Tabor, Josh - Yates, Daniel - Moore, David, The Practice of Statistics, New York: W.H. Freeman, 2015, 5th Edition

Data (Chapter 1)

Evidence of Learning:

Starnes, Daren - Tabor, Josh - Yates, Daniel - Moore, David, The Practice of Statistics, New York: W.H. Freeman, 2015, 5th Edition

Section (1-2) Exercises

Basic Application: p. 41 #38 a

Strategic Thinking: p43 #49b

Basic Application: p.77 R1.7 a, b

Strategic Thinking: p. 77 R1.7 c, p. 76 R.1.3 a, b

Capstone Connection:

Representing data derived from test line equation then can expect values of x given plan and find correlation coefficient and regression y

Modeling With Geometry

Weeks: Week 07

MA.1.03:

HSS MA.3.0

Apply the relationships between 2-D and 3-D objects in modeling situations

Concepts	Skills
•1. Cross section •2. 2-Dimensional objects •3. 3-Dimensional objects •4. Classifying regular polyhedral	•1. Draw 2-D cross-sections of a representation of a 3-D object. •2. Identify three dimensional objects created by rotations of two-dimensional objects •3. Describe 3-D shapes •4. Create 3-D shapes 5- determine the shape of top view and side view of any solid 6-determine the solid of any net shape if it folded

Essential Question(s): Why is the world 3-D?

Big Idea:

We use mathematical modeling with a variety of classes of functions to solve applications such as in biology and finance, and to form a basis for the study of mechanics and calculus.

Textbook and Resource Materials:

Math Connections Geometry 2b Section 5.1

See materials in the Unit Materials Folder:

<https://drive.google.com/drive/folders/0B10u0LfTYd6hfIM3QkZnM0JCM1B6bzR5T3F6dktnN0RkdFBjb3RpSXdBLTRuMIVibVZGWW8>

Evidence of Learning:

Problem Set 5.1 BA:p 466 # 1 a, b, ST:466 # 1 c

Assessment in "2D Representations of 3D Objects"

Capstone Connection:

Generate 3-d solids of rotating 2-d designed shapes using sketch up program or any other program (building houses, bridges,.....etc)

2-D/3-D Objects

Weeks: Week 08 - Week 9

MA.1.04:

HSS MA.3.06

Understand similarity and use the concept for scaling to solve problems.

Concepts	Skills
•1. Similarity •2. Transformation (reduction and magnification) •3. Proportionality •4. Parallel lines and proportional parts •5. Angle bisectors	•1. Determine axes of similarity •2. Draw the image of a shape (using reduction and magnification) •3. Solve problems and applications using proportionality theorems •4. Prove the angle bisector theorem •5. Stretching and shrinking angles and areas •6. Stretching and Shrinking volumes •7. Finding the scaling factor (area and volume) •8. Solve problems related to geometric shapes "

Essential Question(s): Why are relationships predictable?

Big Idea:

We use mathematical modeling with a variety of classes of functions to solve applications such as in biology and finance, and to form a basis for the study of mechanics and calculus.

Textbook and Resource Materials:

Geometry Packet in the Unit Materials Folder pages 1 to 25

<https://drive.google.com/drive/folders/0B10u0LfTYd6hfkIIWmNyWHcwRDdRWXJ5T05peldMVFBtemtscTczU1IDaEVvaWo4aTZ6bDg4>

Math Connections 2a 1.10

Math Connections Geometry 2a, 2.9, 2.10

Geometry Packet in the Unit Materials Folder pages 27 to 38

Evidence of Learning:

Geometry Packet Basic Application- page 4, page 16 #7, page 25 bottom chart

Geometry Packet Strategic Thinking Pages 12, 18

Math Connections Geometry 2a, 2.10 BA - #5

Geometry Packet Strategic Thinking page 31 #4, page 36, page 38

Capstone Connection:

Design prototype its dimensions are proportional to the real project using similarity

SEC Topic:

Similarity, scaling

Weeks: Week 10 - Week 11

MA.1.05:

HSS MA.3.10 MA.3.11 MA.3.12 MA.3.13

Apply volume formulas (pyramid, cones, spheres, prisms).

Concepts	Skills
•1. Area •2. Volume •3. Composite figures	•1. Derive the formulas of areas and volumes •2. Determine area and volume of different shapes and figures •3. Determine volume of a composite figures

Essential Question(s): How does what we measure influence how we measure?

Big Idea:

We use mathematical modeling with a variety of classes of functions to solve applications such as in biology and finance, and to form a basis for the study of mechanics and calculus.

Textbook and Resource Materials:

Math Connections 2b 5.3-5.4

See materials in Unit Materials folder:

<https://drive.google.com/drive/folders/0B10u0LfTYd6hfjJHNDFJTGpkVIQ1S040Z2JlOWhRS3JENkFQUzNVVzdZNEM2QINLckU1Mlk>

Evidence of Learning:

Math Connections 5.3 Basic Application #3

Math Connections 5.4 Strategic Thinking #7

Capstone Connection:

Calculate the area and volumes of composite figures (any model of prototype)

Modeling With Functions: Linear And Step Functions

Weeks: Week 12 - Week 14

MA.1.06:

HSS MA.2.0

Create, interpret and analyze functions, particularly linear and step functions that model real-world situations.

Concepts	Skills
•1. Linear functions •2. Piecewise function •3. Absolute-Value Functions •4. STEP function •5. Direct variation •6. Inverse variation •7. Intercepted part •8. Composition of functions •9. Translation of linear function	•1. Given data, create functions that model applied situation. Calculate a least squares regression line, understand its properties, compute and interpret correlation for data •2. Interpret the functions based on the real-world situation they model •3. Analyze the functions and their characteristics. Explain the different properties of the algebraic function and its graph. •4. Identify, interpret and describe real-world step functions •5. Analyze direct and inverse variations based on the real-world situation they model. •6. Graph linear, piecewise, absolute value and step functions •7. Find the inverse of some functions algebraically and graphically and determine if the inverse is a function. (Also: Distinguish between "inverse variation" and "inverse function.") •8. Solve real life application on functions

Essential Question(s): •Why is modeling an indispensable tool?

•Why do we represent phenomena in multiple ways?

•How does data help make sense of the world around us?

Big Idea:

We use mathematical modeling with a variety of classes of functions to solve applications such as in biology and finance, and to form a basis for the study of mechanics and calculus.

Textbook and Resource Materials:

Math Connections 1A 3.4 - 3.6, 4.4 - 4.5

Math Connections 1B 6.1 - 6.4, 6.6

<https://www.youtube.com/watch?v=npBg0Ynv4D0>

See material in Unit Materials folder

Evidence of Learning:

Quiz : <https://drive.google.com/drive/folders/0B9KK7ZY6QeS3NGZVd25rbWhmS2c>

Math Connections 1a

Basic Application Problem Set 4.5 #1

Strategic Thinking Problem Set 4.5 #6

Math Connections 1b

Basic Application Problem Set 6.1 #2, Problem Set 6.3 # 3, Problem Set 6.4 #6 a through c, Problem set 6.6 # 2 (any of them)

Strategic Thinking Problem Set 6.1 #8, Problem Set 6.4 #6 d, Problem Set 6.6 #6c

Capstone Connection:

Drive relation between variables in capstone, then determine the type of relation